Final Portfolio

Seminar in Algebra

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CHAPTER 1

SUMMARY PAGE

1.1 Introduction

From a fundamental perspective, we study abstract algebra to explore structure. In that, by studying structure, we can discover shared properties between abstract systems, while still operating under rules and axioms. Furthermore, understanding structure allows us to filter, dissect, and categorize seemingly chaotic mathematical spaces. We use homomorphisms to project complex algebraic structures down to simpler ones, revealing the underlying mechanics of the system. The kernel of a homomorphism acts as an elemental filter, letting us mod out specific information, while the image captures the surviving structure. Once a system is reduced, we use isomorphisms as our main tool for classification, to identify when two algebraic structures behave in the same way structurally, even if they appear vastly different on the surface. Therefore, by breaking down complex systems into their fundamental components with the properties of homomorphisms and classifying them through isomorphisms, we can build up a stronger understanding of algebraic behavior.

Thus, in this portfolio, I will explore how homomorphisms and isomorphisms act as the primary mechanisms for reducing and classifying algebraic systems. Through selected proofs spanning groups, rings, and fields, I will demonstrate how these mapping tools allow us to identify and remove differences, collapse complex sets into manageable factor spaces, and ultimately uncover the fundamental building blocks of algebra.

1.2 Homomorphisms & Their Kernels

The definition of a homomorphism provides a mapping between two algebraic structures, while also preserving element-wise operations. Thus, we use this mapping as a means to transform rigid or overly complex structures into manageable parts while maintaining the structure that we want. Viewed through this angle, the homomorphism provides the pathway and its kernel, defined as $\{a \in R \mid \varphi(a) = 0'\}$, removes unwanted aspects to yield a product we can work with. An example of this filtering functionality can be seen with the compare and contrast proofs from Proof 1 in the portfolio. In these proofs, we look at an arbitrary group homomorphism mapping some group G to another group G'. Even without knowing anything about the specific elements inside these groups, the mapping's kernel partitions the domain perfectly, preserving enough structure to tell us how their sizes relate.

The non-compromising nature of this filtering process can be best seen when applied to structures that cannot be broken down, such as fields. Proof 17 from the portfolio shows that any homomorphism from a field to a ring is either one-to-one, or maps everything to $0'$. This happens because the kernel of any ring homomorphism must be an ideal. Since

a field cannot have any proper non-trivial ideals, its kernel is restricted to exactly two options: the trivial ideal $\{0\}$ or the entire field itself. In other words, a field's structure forces the kernel to either preserve everything from the field (a one-to-one map), or discard all of it to zero. Thus, we see that every homomorphism and its kernel must abide by strict rules imposed by the structural limitations of the domain it is applied to.

1.3 Quotients & Factor Spaces

Since we defined homomorphisms and their kernels as structural filters, it makes sense to explore the spaces left behind after filtering. Hence, we arrive at the concept of quotient structures. Now, to make any factor group or factor ring, we can only quotient by either normal subgroups or ideals. This restriction is required to ensure that the component-wise operations remain well-defined. In Proof 3 from the portfolio, we explore this well-defined property by proving that if the left cosets of a subgroup H are the same as its right cosets, then $g^{-1}hg \in H$ for all $g \in G$ and $h \in H$. This normality condition is thus the exact structural prerequisite needed to collapse a group into a valid factor group.

Once this normality has been established, we can map out how factor spaces operate. One example of this can be seen in Proof 8 of my portfolio, where I built a coset table for the subgroup $\{p_0, p_2\}$ within the dihedral group D_4. Essentially, by reducing the symmetries of a square by this subgroup, we are left with a new quotient group of order 4. To find whether this new structure operates like the Klein-4 group or $\mathbb{Z}_4$, I showed that each element of the coset behaves exactly like the elements of the Klein-4 group. This exercise highlights the importance of quotients: we can take a rigid and non-abelian group like D_4, and fold it into a smaller, abelian system.

This same logic applies to rings, where the structural prerequisite shifts from normal subgroups to ideals. In Proof 12, I compare two different ways ideals interact with ring structures. I show how a ring homomorphism maps an ideal N to a new ideal $\varphi[N]$, and I contrast this with proving the set $I_a = \{x \in R \mid ax = 0\}$ forms an ideal within R. By identifying these ideals, we establish the framework needed to mod out specific elements and collapse a large ring into a smaller factor ring.

1.4 Isomorphisms

After using kernels and ideals to collapse our structures into factor spaces, we need a way to identify and relate the new structure to what we know. Thus, we use isomorphisms as our main tool for classification. While a homomorphism shows how two structures relate, an isomorphism tells us when two structures are fundamentally identical.

We see this classification process explicitly in Proof 4. Here, we examine the additive group of functions mapping R into R, denoted as F, and its subgroup of constant functions, K. By finding a subgroup isomorphic to the factor group F/K, we identify the essential structure that remains when we strip away all constant behavior. The isomorphism bridges the gap between an abstract quotient space and a recognizable subgroup.

Furthermore, isomorphisms help us categorize the boundaries of algebraic behavior. In Proof 13, I build on the groundwork laid in Proof 15 by showing that a factor ring of a field is either the trivial zero ring or is isomorphic to the field itself. Using the same fact that a field has no proper non-trivial ideals, we can force the structure of the factor ring to either be reduced to nothing, or remain identical to the original field.

1.5 Field Extensions

In the earlier topics of the course, homomorphisms and quotients were primarily used to break structures down. However, these same mechanisms can be used to construct new systems. Instead of modding out by an ideal to simplify a space, we can mod out by a maximal ideal generated by an irreducible polynomial to build a larger space where new solutions exist.

This constructive property is the focus of my Major Theorem selection in Proof 9. The theorem guarantees that for any field F and any nonconstant polynomial $f(x)$, there exists an extension field E containing a root of that polynomial. By constructing the quotient ring $F[x]/\langle p(x)\rangle$, where $p(x)$ is an irreducible factor, we produce a space where the previously unsolvable polynomial now has a root. We use the mechanics of polynomials and ideals to build a new structure entirely.

Once these new extension fields are constructed, we measure and classify their internal architecture. Proof 11 investigates the field extension $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ over $\mathbb{Q}$. By determining its degree and finding a basis, we apply the tools of linear algebra to understand the size of this algebraic extension. Just as we measured the size of a homomorphic image using divisors in Proof 1, we measure the complexity of our new fields through their bases and degrees.

1.6 Conclusion

It can be tempting to see abstract algebra as a disjointed collection of axioms, groups, rings, and fields. However, as this portfolio demonstrates, the study of algebra is unified by the exploration of structure.

Through the proofs curated in this portfolio, a clear narrative emerges. We use homomorphisms as filters to navigate between systems, relying on kernels to isolate the properties we want. We use normal subgroups and ideals to collapse unwieldy structures into manageable factor spaces. We deploy isomorphisms to classify these resulting spaces, proving when seemingly different systems are structurally identical. Finally, we use irreducible polynomials and quotient rings to build extension fields that provide solutions.

Looking back on the progression from basic group definitions to field extensions, the underlying search for structure remains the same. In five years from now, I may not be able to remember how to construct a dihedral coset table, or find the exact prime ideal of a polynomial ring. Yet, the foundational understanding of how to analyze, deconstruct, and categorize mathematical spaces will remain.

CHAPTER 2

PORTFOLIO ITEMS

1. (**Compare and Contrast**) These two proofs are focused on exploring the same concept of the homomorphisms's image, but from opposite ends of the mapping. They both hinge on Lagrange's Theorem to enforce restrictions upon the size of the image, but they emphasize different mechanics of how a homomorphism acts as a structural filter.

   This proof works by using Lagrange's Theorem to relate the size of the image to the size of the domain. My strategy was to identify a subset of the domain that partitions it perfectly—the cosets of the kernel—and link those to the image. The difficulty arose when trying to rigorously connect the cosets (G/H) to the image ($\varphi[G]$). Theorem 13.15 saved me here, as it establishes the necessary one-to-one correspondence between them. By showing that the kernel filters the domain into cosets, we see that the structure of the domain forces the size of the image to be a divisor of $|G|$.

   In contrast, the second proof is entirely focused on the codomain (G'), and not on the kernel. Thus, I knew that I only needed to prove that the image $\varphi[G]$ is a valid subgroup of G'. Once I established this with using the fundamental properties of homomorphisms, the rest fell into place via Lagrange's Theorem. Then, because the image satisfies the group axioms, it must adhere to the structural limitations of the parent group, meaning its order must divide $|G'|$. Through these proofs, we see that our mapping filter is so rigid that the resulting image is both locked into the structural boundaries of both its starting point and its destination.

   Problem. Let $\varphi : G \to G'$ be a group homomorphism. Show that if $|G|$ is finite, then $|\varphi[G]|$ is finite and is a divisor of $|G|$.

   Proof. Let $\varphi : G \to G'$ be a group homomorphism, and let $H = \ker(\varphi)$. Our goal is to show that the order of the preimage of G, $|\varphi[G]|$, is finite, and a divisor of $|G|$, given G is finite. Theorem 13.15 tells us that $aH = \varphi^{-1}[\{\varphi(a)\}]$ maps a single coset to the same single element $\varphi(a)$ in the image. This establishes a one-to-one correspondence between the set of all cosets (G/H) and the set of all images, $\varphi[G]$. Because there is a one-to-one correspondence, the two sets must have equal size: $|G/H| = |\varphi[G]|$. Then, by Lagrange's Theorem, since G is finite, the number of cosets (G/H) must be a finite number that divides the order of the group, $|G|$. Since $|G/H| = |\varphi[G]|$, it follows that $|\varphi[G]|$ must also be a finite number that divides $|G|$. $\square$

   Other proof on the next page.

Problem. Let $\varphi : G \to G'$ be a group homomorphism. Show that if $|G'|$ is finite, then $|\varphi[G]|$ is finite and is a divisor of $|G'|$.

Proof. From Theorem 13.12 (3) (the fundamental properties of homomorphisms), if H is a subgroup of G, then $\varphi[H]$ is a subgroup of G'. Now, let $H = G$. It is certainly true that G is a subgroup of G, so $\varphi[G]$ is a subgroup of G'. Then, by Lagrange's Theorem, since $|G'|$ is finite, the order of its subgroup, $|\varphi[G]|$ is also a finite number that divides $|G'|$. □

2. This proof is concerned with transitivity through composite maps. I showed that for any three groups connected through two homomorphisms, the composite mapping is also a homomorphism. These types of direct computation proofs are my favorite; by just leveraging definitions, we are able to draw connections between group structures with logical steps that are easily auditable. Consequentially, I found that the structure-preserving property holds up even when we chain multiple systems together, allowing us to build larger mathematical arguments from smaller, verified blocks.

Problem. Show that if G, G', and G'' are groups and if $\varphi : G \to G'$ and $\gamma : G' \to G''$ are homomorphisms, then the composite map $\gamma\varphi : G \to G''$ is a homomorphism.

Proof. Let $a, b \in G$. Then,

$$\begin{aligned}
(\gamma\varphi)(ab) &= \gamma(\varphi(ab)) && (2.1)\\
&= \gamma(\varphi(a)\varphi(b)) && (2.2)\\
&= \gamma(\varphi(a))\gamma(\varphi(b)) && (2.3)\\
&= (\gamma\varphi)(a)(\gamma\varphi)(b). && (2.4)
\end{aligned}$$

Equations (1) and (4) are from the definition of a composite map, and (2) and (3) are from the homomorphic properties of φ and γ, respectively. Therefore, we have shown that the composite map is a homomorphism. □

3. This proof deals with the specific conditions required for left and right cosets to form identical partitions. My goal was to prove if the partitions are the same, then the element $g^{-1}hg$ must belong to the subgroup. Writing this required a shift in perspective, moving from thinking about individual element operations to thinking about set equality ($gH = Hg$). This proof shows that the property of normality is the structural requirement that allows us to identify factor group candidates from subgroups.

Problem. Let H be a subgroup of a group G. Prove that if the partition of G into left cosets of H is the same as the partition into right cosets of H, then $g^{-1}hg \in H$ for all $g \in G$ and all $h \in H$.

Proof. The statement "If the partition of G into left cosets of H is the same as the partition into right cosets of H," implies $gH = Hg$ for all $g \in G$. Now, let $g \in G$ and $h \in H$. Our goal is to show that $g^{-1}hg \in H$. Consider the element hg. Since $h \in H$, $hg \in Hg$. Because $Hg = gH$, it must be that $hg \in gH$. By definition of left coset, $hg \in gH$ means that $hg = gh'$ for some $h' \in H$. So, we just solve for this h'. We start by multiplying both sides on the left by g^{-1}:

$$\begin{aligned} g^{-1}(hg) &= g^{-1}(gh') \\ g^{-1}hg &= (g^{-1}g)h' \\ g^{-1}hg &= h'. \end{aligned}$$

Since $h' \in H$, we have shown that $g^{-1}hg \in H$. □

4. For this proof, I moved away from theorems to an application involving groups of functions. My goal was to identify the structure of the factor group F/K by finding a simpler group it is isomorphic to. To do this, I used the Fundamental Homomorphism Theorem: I defined a map that eliminates the constant functions (the kernel), and then used the theorem to show that the quotient group is isomorphic to the image. So, we took a complex structure (F/K) and used a homomorphism to classify it as being structurally identical to the group of functions rooted at zero.

Problem. Let K be the subgroup of F consisting of the constant functions, where F is the additive group of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Find a subgroup of F to which F/K is isomorphic.

Proof. Define a homomorphism $\varphi : F \to F$ by $\varphi(f) = f(x) - f(0)$. The kernel of this map consists of all functions where $f(x) - f(0) = 0$, which implies $f(x)$ is constant, so $\ker(\varphi) = K$. The image of the map, $\varphi[F]$, is the set of all functions that evaluate to 0 at $x = 0$. Since $\ker(\varphi) = K$, the Fundamental Homomorphism Theorem states that $F/\ker(\varphi) \cong \varphi[F]$. □

5. (Most Improved) I first solved this problem on the first in-class exam, and did quite poorly, as I forgot the definition of the kernel, and thought the kernel mapped elements to the *multiplicative* identity of R' instead of the additive. So, in my first approach, I botched the kernel definition, and did not use homomorphic property altogether. However, I was able to provide a general "idea" of what I needed to do: Show that $ab \notin \ker(\varphi)$ by using the fact that $a, b \notin \ker(\varphi)$ and R' has no zero divisors. I just did not *show* the process and I went astray with the kernel definition.

Thus, we arrive at the revised proof, where the *correct* definition of the kernel is front and center, in the second sentence. Then, I set $\varphi(ab) = 0'$ to lead to a contradiction, thus proving that $ab \notin \ker(\varphi)$. I did this by using the homomorphic property of φ to establish an equality between $\varphi(ab)$ and $\varphi(a)\varphi(b)$. Then, I dug a level deeper and pointed out that neither $\varphi(a)$ nor $\varphi(b)$ can be equal to $0'$ by the problem statement. Then, I used R''s property as an integral domain, and pointed out that it has no zero divisors, so the product $\varphi(a)\varphi(b) \neq 0'$. I finished off by connecting the dots: since $\varphi(a)\varphi(b) \neq 0$, then $\varphi(ab) \neq 0$, so $ab \notin \ker(\varphi)$.

Problem. Let R and R' be rings, let $\varphi : R \to R'$ be a homomorphism, and let $a, b \in R$. Prove that if R' is an integral domain and $a, b \notin \ker(\varphi)$, then $ab \notin \ker(\varphi)$.

Proof. Let φ be a homomorphism that exists between rings R and R', defined as $\varphi : R \to R'$, and let $a, b \in R$. Recall that $\ker(\varphi) = \{r \in R \mid \varphi(r) = 0'\}$, where $0'$ is the additive identity of R'. Now, suppose that $a \notin \ker(\varphi)$, $b \notin \ker(\varphi)$, and R' is an integral domain. To show that $ab \notin \ker(\varphi)$, suppose to the contrary that $ab \in \ker(\varphi)$ instead. This means $\varphi(ab) = 0'$. Since φ is a homomorphism, we can expand the left side to get $\varphi(ab) = \varphi(a)\varphi(b) = 0'$. However, this cannot be true, as $\varphi(a)$ and $\varphi(b)$ cannot equal $0'$, and their product cannot equal $0'$ since R' is an integral domain and has no zero divisors. Thus, it must be that $ab \notin \ker(\varphi)$. $\square$

6. In this proof, my goal was to find a zero in $\mathbb{Z}_p[x]$ that matched any $a \in \mathbb{Z}_p[x]$. I did this by first noting that a polynomial is reducible if it is greater than or equal to 2, and it has a zero. Then, I used a corollary to Fermat's Little Theorem to show that for the x^p term, since it is equivalent to x, we can set $x = -a$ to get $-a + a \equiv 0 \pmod{p}$. Thus, since a zero was found, the polynomial is reducible.

Problem. Show that for p a prime, the polynomial $x^p + a$ in $\mathbb{Z}_p[x]$ is not irreducible for any $a \in \mathbb{Z}_p$.

Proof. Since p is prime, the degree of our polynomial is $p \geq 2$. We know that a polynomial of degree 2 or higher is not irreducible if it has a zero in the field. So, we plan to find a zero. Remember that Corollary 20.2 tells us that $x^p \equiv x \pmod{p}$ for any $x \in \mathbb{Z}$. Thus, evaluating the polynomial at $x = -a$ gives

$$(-a)^p + a \equiv -a + a \equiv 0 \pmod{p}.$$

Therefore, we have shown that $-a$ is a zero, meaning the polynomial can be factored and is not irreducible. □

7. This proof dealt with using the properties of integral domains, to show that $D[x]$ is also an integral domain. First, since the coefficients of $D[x]$ reside in D, I used Theorem 22.2 to check off all but one property of integral domains: zero divisors. So, I just needed to find two nonzero elements in $D[x]$ and show that their product does not equal 0. So, I let $f(x)$ be of degree n, and $g(x)$ of degree m and multiplied them together, with at least one element in each being nonzero, a_n and b_m respectively. These were what I multiplied together to show that their product is nonzero, using D's no zero divisors in the process.

Problem. Prove that if D is an integral domain, then $D[x]$ is an integral domain.

Proof. Let D be an integral domain. We will show that $D[x]$ satisfies all integral domain axioms. From Theorem 22.2, since D is an integral domain, it is also a commutative ring with unity $1 \neq 0$. Hence, $D[x]$ is also a commutative ring with unity $1 \neq 0$. So, we are left with showing that $D[x]$ has no zero divisors. Let $f(x)$ and $g(x)$ be nonzero polynomials in $D[x]$, where $f(x)$ has degree n, and $g(x)$ has degree m. Thus, we have

$$f(x) = a_n x^n + \cdots + a_1 x + a_0 \quad \text{and} \quad g(x) = b_m x^m + \cdots + b_1 x + b_0,$$

where $a_n \neq 0$ and $b_m \neq 0$. Now, consider the product $f(x)g(x)$. The term with the highest degree in $f(x)g(x)$ is $a_n b_m x^{n+m}$. Since D is an integral domain, and $a_n, b_m \neq 0$, we know their product $a_n b_m \neq 0$. Therefore, $f(x)g(x)$ is nonzero. Thus, $D[x]$ has no 0 divisors, and is an integral domain. □

8. (A High Point) In this problem, I used the group of symmetries of the square, D_4 in combination with its 8 permutations: ρ_i, μ_i, δ_i for rotations, mirror images, and diagonal flips respectively. I also used the Klein 4-group, which is the simplest abelian group that is not cyclic. Then, I found the left cosets by multiplying $\{\rho_0, \rho_2\}$ by each permutation. This gives an isomorphism between D_4/H and V because they have the exact same structure. I am particularly proud of this problem because of the significance of the two groups in the same problem. Throughout the semester, we had talked extensively about the Klein 4-group, and we had marveled at the all the symmetries of the D_4 group. Thus, due to the relevance of these two groups (and the beauty of the tables next to each other), I figured I must include it.

Problem. Build a table in the order exhibited by the left cosets of the subgroup $\{\rho_0, \rho_2\}$ of the group D_4. Do you seem to get a coset group of order 4? If so, is it isomorphic to $\mathbb{Z}_4$ or the Klein 4-group V?

Solution. The multiplication table on the left is made up of left cosets for D_4. The subgroup $H = \{\rho_0, \rho_2\}$ partitions D_4 into the four left cosets. Namely,

$$\{\rho_0, \rho_2\},\ \{\rho_1, \rho_3\},\ \{\mu_1, \mu_2\},\ \{\delta_1, \delta_2\}.$$

Each colored block corresponds to one of these cosets. The table on the right is the multiplication table for the Klein 4-group. Notice that

$$\{\rho_0, \rho_2\} = e,\ \{\rho_1, \rho_3\} = a,\ \{\mu_1, \mu_2\} = b,\ \{\delta_1, \delta_2\} = c.$$

Hence, $D_4/H \cong V$.

	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_0	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_2	ρ_2	ρ_0	ρ_3	ρ_1	μ_2	μ_1	δ_2	δ_1
ρ_1	ρ_1	ρ_3	ρ_2	ρ_0	δ_1	δ_2	μ_2	μ_1
ρ_3	ρ_3	ρ_1	ρ_0	ρ_2	δ_2	δ_1	μ_1	μ_2
μ_1	μ_1	μ_2	δ_2	δ_1	ρ_0	ρ_2	ρ_3	ρ_1
μ_2	μ_2	μ_1	δ_1	δ_2	ρ_2	ρ_0	ρ_1	ρ_3
δ_1	δ_1	δ_2	μ_1	μ_2	ρ_1	ρ_3	ρ_0	ρ_2
δ_2	δ_2	δ_1	μ_2	μ_1	ρ_3	ρ_1	ρ_2	ρ_0

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

9. (**Major Theorem**) This is Kronecker's Theorem, and referenced throughout *A First Course in Abstract Algebra* by John Fraleigh and Victor Katz as the "basic goal" they are trying to accomplish. That is, "to show that any nonconstant polynomial $f(x)$ in $F[x]$ has a zero in some field E containing F" (pg. 250). Thus, I thought it appropriate to include it as a Major Theorem in this portfolio. The proof itself involves irreducible polynomials, factor rings, an isomorphism, cosets, and the evaluation homomorphism. It highlights the importance of using maps to achieve structure by embedding our original field into a factor ring where the missing root emerges.

For this proof, I thought that defining $\alpha = x + \langle p \rangle$ was a very clever technique. By treating the indeterminate x itself as the representative of a coset, we force the polynomial to evaluate to its own ideal, which acts as the zero element in the new field.

Problem. Let F be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Then there exists an extension field E of F and an $\alpha \in E$ such that $f(\alpha) = 0$.

Proof. Let F be a field, and let $f(x)$ be a nonconstant polynomial in $F[x]$. Then, by Theorem 23.20, we can write this $f(x)$ as a product of irreducible polynomials over F. Let $p(x)$ be a polynomial in this factorization with $\deg(p(x)) > 1$. Then, by Theorem 27.25, since this polynomial is irreducible over F, $\langle p(x) \rangle$ is a maximal ideal of $F[x]$. Hence, $E = F[x]/\langle p(x) \rangle$ is a field. Now, to show that F is a subfield of E, let $\psi : F \to F[x]/\langle p(x) \rangle$ be given by $a \mapsto a + \langle p(x) \rangle$ for $a \in F$. To show that this map is a homomorphism, let $a, b \in F$. We check the operations of addition and multiplication:

$$\psi(a+b) = (a+b) + \langle p(x) \rangle = (a + \langle p(x) \rangle) + (b + \langle p(x) \rangle) = \psi(a) + \psi(b)$$

$$\psi(ab) = (ab) + \langle p(x) \rangle = (a + \langle p(x) \rangle)(b + \langle p(x) \rangle) = \psi(a)\psi(b)$$

Thus, ψ is a homomorphism.

Next, we show that ψ is one-to-one so that we can identify F with a subfield of E. Suppose $\psi(a) = \psi(b)$ for some $a, b \in F$. Then $a + \langle p(x) \rangle = b + \langle p(x) \rangle$, which implies $a - b \in \langle p(x) \rangle$. Because $a, b \in F$, their difference $a - b$ is a constant polynomial. However, the ideal $\langle p(x) \rangle$ only consists of multiples of $p(x)$. Since $\deg(p(x)) > 1$, the only constant polynomial in $\langle p(x) \rangle$ is 0. Therefore, $a - b = 0$, meaning $a = b$. This shows ψ is one-to-one, and that there is an isomorphic copy of F in E: $\{a + \langle p(x) \rangle \mid a \in F\}$.

Finally, let $\alpha = x + \langle p(x) \rangle \in E$. We must show that α is a root of $f(x)$. Since $p(x)$ is a factor of $f(x)$, we can write $f(x) = p(x)q(x)$ for some $q(x) \in F[x]$. Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n$. We use the evaluation homomorphism $\varphi_\alpha : F[x] \to E$ to find

$$\varphi_\alpha(p(x)) = a_0 + a_1(x + \langle p(x) \rangle) + \cdots a_n(x + \langle p(x) \rangle)^n$$

We can choose x to be representative of the coset $x + \langle p(x) \rangle$ so that

$$\begin{aligned} p(\alpha) &= (a_0 + a_1 x + \cdots + a_n x^n) + \langle p(x) \rangle \\ p(\alpha) &= p(x) + \langle p(x) \rangle \\ p(\alpha) &= \langle p(x) \rangle. \end{aligned}$$

Thus, $p(\alpha) = \varphi_\alpha(p(x)) = 0$ in E. Therefore, $\alpha \in F[x]/\langle p(x) \rangle$ such that $f(\alpha) = 0$. □

10. I did this proof incorrectly at first, and only realized it when Noah asked me for a hint on how to start. We both knew that we needed to incorporate all information from the problem (e.g., use $n, m \in \mathbb{Z}^+$ to show that $(a+b)^{n+m} = 0$), but we initially didn't know how to incorporate the commutativity part of the problem. Then, he realized that we could use the binomial theorem, which just called for two cases to be worked through.

Problem. An element a for ring R is **nilpotent** if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that if a and b are nilpotent elts of a commutative ring, then $a + b$ is also nilpotent.

Proof. Let $a, b \in R$, a commutative ring. This means there exists some $n, m \in \mathbb{Z}^+$ for which $a^n = 0$ and $b^m = 0$. Our goal is to show that $(a + b)^{n+m} = 0$. Since R is a commutative ring, we can use distributive laws and the binomial theorem to help here. Thus, consider the following:

$$(a + b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^k b^{(n+m)-k}.$$

Note that every element in each term is a multiple of our nilpotent elements, so our goal is to show that each of these elements are 0 for any value of k. Thus, consider these two cases:

- $k \geq n$: If $k \geq n$, then $a^k = a^n a^{k-n} = 0a^{k-n} = 0$, so the whole term is 0.
- $k < n$: Since $k < n$, it follows that $-k > -n$. Therefore, $(n + m) - k > n + m - n = m$ (because $\langle R, + \rangle$ is an abelian group). Since the exponent of b is greater than m, $b^{(n+m)-k} = 0$.

Therefore, we have shown that $(a + b)^{n+m} = 0$. □

11. (**High Point**) I am very proud of this solution, as solving the problem outright is not possible, and requires building another finite extension to use the Tower Theorem (Theorem 31.4). So, I took the given field extension $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ and found another field for which it was also an extension for: $\mathbb{Q}(\sqrt[3]{2})$. This new field allowed me to build the tower, by also being an extension of $\mathbb{Q}$. Thus, I had two bases, and 2 field extensions, so I could use the Tower Theorem and finally answer the main problem. This solution was a bit tedious, but the reward from checking the edge cases was worth it.

Problem. Find the degree and basis for the field extension $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ over $\mathbb{Q}$.

Solution. We start by showing that $\sqrt{2} \notin \mathbb{Q}(\sqrt[3]{2})$ because $\deg(\sqrt{2}, \mathbb{Q}) = 2$ and 2 is not a divisor of $3 = \deg(\sqrt[3]{2}, \mathbb{Q})$. Therefore, we have $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ over $\mathbb{Q}(\sqrt[3]{2})$ with $f(x) = x^2 - 2$. Thus,

$$[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}(\sqrt[3]{2})] = \deg(f(x)) = 2.$$

Then, we find $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$ has degree $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ for $g(x) = x^3 - 2$. Then, we use Theorem 30.23 to produce bases $\{1, \sqrt{2}\}$ for $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ over $\mathbb{Q}(\sqrt[3]{2})$, and $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$ for $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$. By Theorem 31.4, we find

$$[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}(\sqrt[3]{2})][\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 6,$$

and we get a basis (from the proof of Theorem 31.4):

$$\{1, \sqrt{2}, \sqrt[3]{2}, \sqrt{2}\sqrt[3]{2}, (\sqrt[3]{2})^2, \sqrt{2}(\sqrt[3]{2})^2\}$$

for $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ over $\mathbb{Q}$. Finally, we note that $f(x)$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$ by Theorem 23.10, and $g(x)$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion for $p = 2$.

12. (**Compare and Contrast**) These two proofs deal with finding ideals of a ring through different methods. In both proofs, I needed to show two things: the ideal in question is a subring, and that when elements in R are multiplied by the ideal, that they stay within the ideal. The first accomplishes this by using Theorem 26.3, which gives $\varphi[N]$ as a subring of $\varphi[R]$, since N is a subring of R. Then, I started with arbitrary $x \in \varphi[R]$ and $y \in \varphi[N]$, and used N's ideal property along with the homomorphism property of φ to show that xy and yx are contained in $\varphi[N]$. Then, for the second proof, I proved that I_a is a subring of R without using the commutativity of R, but by purely just using the properties of rings and the definition of I_a. Proving the second property of ideals did require using commutativity, as showing $rx \in I_a$ would be impossible without it. Thus, even though the technique used for both of these proofs are distinct in nature, it shows that we can get ideals of rings through maps, or by the structure of the ideal itself.

Problem. Let $\varphi : R \to R'$ be a ring homomorphism and let N be an ideal of R. Show that $\varphi[N]$ is an ideal of $\varphi[R]$.

Proof. To prove that $\varphi[N]$ is an ideal of $\varphi[R]$, we must show that $\varphi[N]$ is a subring, and for an $x \in \varphi[R]$, that $x\varphi[N] \subseteq \varphi[N]$ and $\varphi[N]x \subseteq \varphi[N]$. Since N is an ideal of R, N is a subring of R. Then, by Theorem 26.3, we know that $\varphi[N]$ is a subring of $\varphi[R]$ as well. To show the second ideal property, let $x \in \varphi[R]$ and $y \in \varphi[N]$. By the definition of image, there exists some $r \in R$ such that $x = \varphi(r)$ and some $n \in N$ such that $y = \varphi(n)$. Since N is an ideal of R, we get $rn \in N$ and $nr \in N$. Therefore, we find:

$$\begin{aligned} xy &= \varphi(r)\varphi(n) = \varphi(rn) \in \varphi[N] \\ yx &= \varphi(n)\varphi(r) = \varphi(nr) \in \varphi[N]. \end{aligned}$$

Therefore, $\varphi[N]$ is an ideal of $\varphi[R]$. □

Problem. Let R be a commutative ring and let $a \in R$. Show that $I_a = \{x \in R \mid ax = 0\}$ is an ideal of R.

Proof. To prove that I_a is a subring of R, we must show the following:

(1) $0 \in I_a$ because $a \cdot 0 = 0$. If $b, c \in I_a$ then $a(b - c) = ab - ac = 0 - 0 = 0$ by the distributive laws of R and the definition of I_a. Thus, $b - c \in I_a$.

(2) If $b, c \in I_a$ then $a(bc) = (ab)c = 0 \cdot c = 0$ by the associativity from R and the definition of I_a. Thus, $bc \in I_a$.

Therefore, I_a is a subring of R. Thus, we need only show that $xr \in I_a$ and $rx \in I_a$ for $x \in I_a$ and $r \in R$. Thus, $a(xr) = (ax)r = 0 \cdot r = 0$. So, $xr \in I_a$. Now, since R is a commutative ring, we can rearrange $a(rx) = a(xr) = 0$. So, $rx \in I_a$. Therefore, we have shown that I_a is an ideal of R. □

13. (**Touchstone**) This proof is beautiful because it highlights how rigid fields are. That is, unlike rings or groups, which can be broken down into smaller factor structures through ideals or normal subgroups, fields cannot. Because a field possesses no nontrivial ideals, applying a map only yields two outcomes: we either compress the whole structure into the trivial ring, or we preserve it through an isomorphic copy. Thus, fields cannot be structurally broken down without collapsing entirely.

Problem. Show that a factor ring of a field is either the trivial (zero) ring of one element or is isomorphic to the field.

Proof. Let F be a field with ideal N. Since F is a field and N is an ideal, either $N = F$ or $N = \{0\}$. If $N = F$, then we know from Exercise 18, Section 26, that for all $a \in F$, $\varphi(a) = 0'$. So, the factor ring F/F consists of only one element (the identity element), so it is the trivial ring. Conversely, if $N = \{0\}$, define a mapping $\varphi : F \to F/\{0\}$ by $x + \{0\}$ for $x \in F$. This mapping is surjective by definition. It is also injective by Exercise 18, Section 26. Thus, φ is a bijective homomorphism, meaning it is an isomorphism. Therefore, $F/\{0\} \cong F$. $\square$

14. (**Touchstone**) I like this proof because it is so elegant. In it, I essentially just link theorems together to make a conclusion. Reading the proof statement basically tells you exactly what the theorems are stating without needing to know what the theorems actually say.

Problem. Let R be a finite commutative ring with unity. Show that every prime ideal in R is a maximal ideal.

Proof. Let N be a prime ideal in R. Since R is finite, the number of cosets in R/N must also be finite. Because N is a prime ideal, R/N is a finite integral domain by Theorem 27.15. Then, by Theorem 19.11, R/N is a field. Hence, N is a maximal ideal by Theorem 27.9. $\square$

15. (**Major Theorem**) This proof is included as a major theorem because we use it constantly under the hood for many of the later proofs in the sequence. There are also a lot of elements in this proof that we have used throughout both semesters. That is, it involves a subring proof, an isomorphism proof, and computing a function composition. Each step in the proof heavily relies upon the homomorphic properties of φ to establish properties. This really highlights the imperative nature of ring homomorphisms, and how integral they are to anything we do in abstract algebra.

Problem. (Fundamental Homomorphism Theorem for Rings) Let $\varphi : R \to R'$ be a ring homomorphism with kernel N. Then $\varphi[R]$ is a ring, and the map $\mu : R/N \to \varphi[R]$ given by $\mu(x+N) = \varphi(x)$ is an isomorphism. If $\gamma : R \to R/N$ is the homomorphism given by $\gamma(x) = x+N$, then for each $x \in R$, we have $\varphi(x) = \mu\gamma(x)$.

Proof. To show $\varphi[R]$ is a subring of R', we must verify the following:

- Since φ is a ring homomorphism, then by Theorem 26.3, it must map the additive identity of R to the additive identity of R'. Thus, $\varphi(0) = 0'$, which means $0' \in \varphi[R]$.
- Let $a, b \in \varphi[R]$. This means there exists $r_1, r_2 \in R$ such that $\varphi(r_1) = a$ and $\varphi(r_2) = b$. Then, $a - b = \varphi(r_1) - \varphi(r_2) = \varphi(r_1 - r_2)$. Since $r_1 - r_2 \in R$, it follows that $a - b \in \varphi[R]$.
- Using the same r_1, r_2 as before: $ab = \varphi(r_1)\varphi(r_2) = \varphi(r_1 r_2)$. Since $r_1 r_2 \in R$, it follows that $ab \in \varphi[R]$.

Now, we must show that μ is well-defined and injective. Thus, suppose $x+N = y+N$. Then, $x - y \in N$. Since $N = \ker(\varphi)$, this implies $\varphi(x-y) = 0$ or $\varphi(x) - \varphi(y) = 0$. Thus, $\varphi(x) = \varphi(y)$ implies $\varphi(x+N) = \varphi(y+N)$. This argument also works in reverse to show that μ is one-to-one: if $\mu(x+N) = \mu(y+N)$, then $\varphi(x) = \varphi(y)$, meaning $x - y \in \ker(\varphi)$, so $x+N = y+N$. Then, we show μ is a ring homomorphism:

$$\begin{aligned}
\mu((x+N)(y+N)) &= \mu(xy+N) && \text{By definition of multi. in } R/N \\
&= \varphi(xy) && \text{By definition of } \mu \\
&= \varphi(x)\varphi(y) && \text{Because } \varphi \text{ is a ring homom.}
\end{aligned}$$

Thus, μ is a ring homomorphism. To complete the proof that μ is an isomorphism, we must show that μ is onto. By definition, the codomain of μ is the image $\varphi[R]$. For any $y \in \varphi[R]$, there exists an $x \in R$ such that $\varphi(x) = y$. Thus, $\varphi(x+N) = \varphi(x) = y$. Finally, let $\gamma : R \to R/N$ be given by $\gamma(x) = x+N$. Then, $\mu\gamma(x) = \mu(x+N) = \varphi(x)$. This holds for all $x \in R$, so $\varphi = \mu\gamma$. $\square$

16. This was the first proof that really introduced us to idempotent elements. I included it in this portfolio because the idea of idempotency comes up time and again throughout the course. The actual proof is by computation, which are also my favorite ones to do. At first, this proof can be a bit tricky, because you need to set up the computation to get the final result. But, if you just follow the definition of closure, you can work your way backward to get to the solution.

Problem. Suppose that $*$ is an associative and commutative binary operation on a set S. Show that $H = \{a \in S \mid a * a = a\}$ is closed under $*$. (The elements of H are **idempotents** of the binary operation $*$.)

Proof. Suppose that $*$ is an associative and commutative binary operation on set S. Let $a, b \in H$ and consider the following:

$$\begin{aligned}
(a * b) * (a * b) &= (b * a) * (a * b) && \text{commutative property,}\\
&= b * (a * a) * b && \text{associative property,}\\
&= b * a * b && \text{definition of } H,\\
&= b * b * a && \text{commutative property,}\\
&= (b * b) * a && \text{associative property,}\\
&= b * a && \text{property of } H,\\
&= a * b && \text{commutativity property.}
\end{aligned}$$

Therefore, H is closed because $a * b \in H$. □

17. This proof is used in Proof 13 because it establishes the fact that any homomorphism from a field to a ring is either one-to-one, or that it maps everything to 0. This solution came from the fact that a field only ever has two ideals: itself, and the trivial ring. So, we have two cases: if the ideal is the field, then every element is in the kernel, so every element gets sent to 0. If the ideal is the trivial ring, then the map is one-to-one by the corollary. Doing this proof was nice because a lot of the heavy lifting was done by the fact that fields have only 2 ideals. With that, I just had to connect the remaining dots.

Problem. Show that each homomorphism from a field to a ring is either one to one or maps everything to 0.

Proof. Let $\varphi : F \to R$ be a ring homomorphism with kernel N. Since F is a field and N is an ideal, we know either $N = F$ or $N = \{0\}$ by Corollary 27.6. If $N = F$, then every element in F is in N. So, for all $a \in F$, $\varphi(a) = 0'$ for $0' \in R$. If $N = \{0\}$, then φ is one-to-one by Corollary 26.6. □

18. (**Most Improved**) This proof was originally on the first algebra exam, and it completely caught me off guard because of the mapping. We had done Exercise 23 of Section 2 as a homework problem, but it was a difficult problem for me, and I did not include it on my exam cheat sheet. So, when this proof came up on the exam, I felt lost. Nevertheless, I did know the 3 things that were needed to prove that a map is an isomorphism, so I listed them, but failed at the onto step, and the homomorphism steps. I didn't think to multiply the complex elements inside the map, so I just tried to produce a matrix from the start.

Now, the revised proof flows much better, and correctly identifies that the map is onto almost purely by definition of how it was set up. Finding that the map was one-to-one can be done by just following the definition of injectivity, and going from there. The final homomorphism step was tricky, but after multiplying the elements, we can get a matrix, and split it in two to demonstrate the homomorphism property.

Problem. Let H be the subset of $M_2(\mathbb{R})$ consisting of all matrices of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{R}$. Exercise 23 of Section 2 shows that H is closed under both matrix addition and matrix multiplication. Show that $\langle \mathbb{C}, \cdot \rangle$ is isomorphic to $\langle H, \cdot \rangle$.

Proof. Let the map $\varphi : \mathbb{C} \to H$ be given by $\varphi(a + bi) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{R}$. To show that any map is an isomorphism, we must show that it is one-to-one, onto, and that it is a homomorphism.

- **One-to-one:** If $\varphi(a+bi) = \varphi(c+di)$ for $c, d \in \mathbb{R}$, then $\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} c & -d \\ d & c \end{bmatrix}$ is only true if $a = c$ and $b = d$. Thus, φ is one-to-one.
- **Onto:** Let $\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \in H$, for $a, b \in \mathbb{R}$. Then, $\varphi(a + bi) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ from the definition of φ. Hence, the mapping is onto, and therefore a bijection
- **Homomorphism:** Consider the computation

$$\varphi((a + bi) \cdot (c + di)) = \varphi((ac - bd) + (ad + bc)i) = \begin{bmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{bmatrix}$$

$$\begin{bmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \varphi(a + bi) \cdot \varphi(c + di).$$

Hence, φ is a homomorphism.

Since we have shown that φ is a bijective homomorphism, then φ is an isomorphism. $\square$